H_0 / H_1 + hypothesis-testing foundations

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- A claim about a population mean is tested using sample evidence.
- Need to state H_0 and H_1, a test statistic and a decision.
- Equality belongs in H_0.

2 DECISION / METHOD

1. Define the population parameter.
2. Put the benchmark/equality in H_0.
3. Let wording determine H_1: different, less, or greater.
4. Assume H_0 to build the reference distribution.
5. Quantify how surprising the sample is.

3 FORMULA

$H_0: \mu = \mu_0$
$H_1: \mu \neq \mu_0, \mu < \mu_0, \text{ or } \mu > \mu_0$

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test statistic =
(estimate-null)/SE_under_H0
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4 FORMULA SHEET VS WHAT I KNOW

Gives: inference SE/distribution lanes.
Still know: hypothesis writing, equality placement, claim direction and contextual conclusion.

5 EXCEL ROUTE

Use cells for μ_0, α, n, $\bar{X}$ and σ/s. Excel computes the statistic; hypotheses must still be written from context.

6 TINY WORKED EXAMPLE

Claim: mean differs from 50. Write $H_0: \mu = 50$ vs $H_1: \mu \neq 50$; this is two-tail.

7 TOP EXAM TRAPS

- Putting equality in H_1.
- Writing hypotheses about $\bar{X}$ instead of population μ.
- Saying H_0 is proved.
- Choosing direction from the sample result rather than the claim.

8 SAY ALOUD

H_0 carries the equality; H_1 carries the claim direction; sample evidence tells me whether to reject H_0.

9 EXIT CHECK

For the claim 'mean is below 12', what are H_0 and H_1?

ONE-GLANCE VISUAL ROUTE

